Causal Disease Intelligence: Architecture and Ethical Framework of the HqA Disease Causation Engine and MNTU Molecular Node Tracer for Computational Health Research

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Abstract—Disease causation is among the most computationally underserved problems in modern medicine. Despite the availability of vast genomic, clinical, molecular, and epidemiological datasets, the computational infrastructure required to systematically identify the causal origins of complex and rare diseases remains fragmented, inaccessible, and poorly integrated. This paper presents the proposed architecture and ethical framework of two complementary computational health tools developed by PTH Meridian: HqA (Hard Questions Answered), a multi-modal disease causation engine integrating causal inference, genomic variant analysis, phenotype ontology matching, and molecular network propagation; and MNTU (Molecular Node Tracer Unified), a biological network analysis platform for tracing disease origination and propagation across protein interaction, metabolic pathway, and gene regulatory networks. The paper defines the computational problem space across six domains: rare disease diagnostic delay, antimicrobial resistance surveillance, health data privacy, Indigenous health equity, the energy-health nexus, and pandemic preparedness. A privacy-preserving architecture integrating federated learning with zero-knowledge proofs is described, enabling multi-institutional disease modeling without patient data sharing. The ethical framework addresses open-source auditability, PIPEDA compliance, limitation disclosure, and the responsibility of computational health tools to serve equity as well as efficiency.

Index Terms—bioinformatics, disease causation mapping, causal inference, genomic variant analysis, molecular network analysis, protein interaction networks, rare disease, antimicrobial resistance, federated learning, zero-knowledge proofs, human phenotype ontology, health equity, PIPEDA, open-source health informatics.

I. Introduction

Modern medicine produces data at unprecedented scale. Whole-genome sequencing costs have fallen from approximately $100 million per genome in 2001 to under $1,000 in 2024 [1]. Electronic health record systems capture longitudinal clinical observations across millions of patients. Protein interaction databases contain hundreds of thousands of experimentally verified interactions [2]. Metabolic and gene regulatory pathway maps span thousands of reactions across the known biology of multiple organisms [3]. And yet: a patient with a rare disease waits an average of four to eight years and consults multiple specialists before receiving a correct diagnosis [8][9][10]. Drug-resistant pathogens spread faster than surveillance tracks them [15]. The signal is present. The computational infrastructure to read it is not.

The central problem is not data scarcity but causal integration. Individual datasets — genomic variants, phenotype records, protein networks, environmental measurements — are each insufficient individually to establish disease causation. The computational challenge is integration: connecting across data modalities, identifying causal rather than correlational relationships, and doing so in a framework that is auditable, privacy-preserving, and accessible to the clinical and research institutions that would use the results.

This paper presents the proposed architecture and ethical framework of two computational tools addressing this challenge. HqA (Hard Questions Answered) is a disease causation engine integrating causal inference methodology with multi-modal biological data sources. MNTU (Molecular Node Tracer Unified) is a companion molecular network analysis platform for tracing disease mechanisms at the biological network level. Both systems are in active development and are published as open-source at github.com/PTHMeridian. This paper documents the design rationale, underlying methodology, and ethical framework — not clinical validation results, which are the subject of planned future work

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pending institutional collaboration.

II. Background and Foundational Methods

A. Causal Inference in Biological Systems

Statistical association between variables does not establish causation. This distinction, foundational to epidemiology and clinical research, was formalized by Pearl in his structural causal model framework [4], which provides mathematical machinery for distinguishing correlational from causal relationships using directed acyclic graphs (DAGs), do-calculus interventional reasoning, and counterfactual analysis. Hernán and Robins [5] extend this framework to epidemiological applications, providing the methodology for causal effect estimation from observational health data. Schölkopf et al. [6] articulate the case for causal representation learning in machine intelligence contexts, demonstrating that models trained on causal rather than purely statistical structure generalize more robustly to distribution shifts — a critical property for clinical deployment contexts where training and deployment populations differ.

HqA is designed to operate within a causal inference framework rather than a purely associative one: the system is intended to generate causal hypotheses — ranked, evidenced explanations of disease origin — rather than correlational associations between symptoms and diagnoses.

B. Molecular Network Biology

The network biology framework, established by Barabási and Oltvai [7] in 2004, treats the cell as a system of interacting components whose collective behavior determines biological function and dysfunction. Disease, in this framework, is not primarily a property of individual genes or proteins but of perturbations in network topology: the disruption of hubs, the disconnection of functional modules, and the propagation of pathogenic signals through interaction networks. Menche et al. [11] demonstrated that disease genes tend to cluster in the human interactome, forming localized disease modules whose proximity to one another correlates with clinical comorbidity. This provides the theoretical foundation for MNTU's network propagation approach to disease mechanism tracing.

C. Phenotype Ontology and Symptom Matching

The Human Phenotype Ontology (HPO) [12], developed by Robinson et al., provides a controlled vocabulary of over 13,000 phenotypic abnormalities with explicit semantic relationships between terms. HPO enables computational comparison of patient phenotype profiles against known disease phenotype signatures, supporting systematic symptom-to-diagnosis matching. Köhler et al. [13] demonstrated that semantic similarity scoring against HPO disease associations significantly outperforms unstructured symptom search in rare disease diagnosis support. HqA's phenotype matching module implements an HPO-based semantic similarity approach as a core component of the diagnostic pathway generation pipeline.

III. HqA: Disease Causation Engine

A. System Overview

HqA is designed as a multi-modal causation mapping platform that ingests heterogeneous biological and clinical data, applies causal inference and network analysis methods, and generates ranked causal hypotheses for disease origin in individual patients or patient cohorts. Fig. 1 illustrates the proposed architecture across three processing layers: data ingestion, analysis, and output.

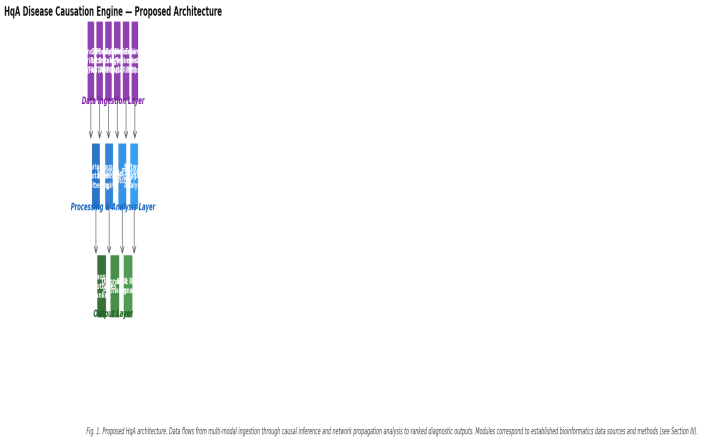

Fig. 1. Proposed HqA architecture. Multi-modal data ingestion feeds causal inference, phenotype matching, and network propagation analysis modules, generating ranked causal hypotheses and diagnostic pathways.

B. Genomic Variant Analysis

Genomic variant analysis in HqA is designed around the clinical variant interpretation pipeline informed by ClinVar [14] — the NCBI repository of genomic variant–disease associations — and the Online Mendelian Inheritance in Man (OMIM) database of monogenic disease gene associations. Variants identified in patient genomic data are annotated, filtered for clinical significance, and mapped to disease associations using ClinVar pathogenicity classifications. Candidate variants are further assessed for network-level impact using the STRING protein interaction database [2], identifying perturbations in the interaction neighborhoods of affected gene products.

C. Rare Disease Application and the Diagnostic Odyssey

The rare disease diagnostic odyssey represents one of the most computationally tractable yet persistently unsolved problems in clinical medicine. Published studies document average diagnostic delays ranging from 4.8 years in U.S. populations [8] to 5.6 years in European cohorts [9], with patients consulting multiple specialists across disciplines before the causal diagnosis is established. The European Organisation for Rare Diseases (EURORDIS) survey [10] of 5,980 patients across 25 countries documented that 25% waited more than 10 years for diagnosis. Fig. 2 illustrates approximate diagnostic delay and specialist consultation burden across rare disease categories.

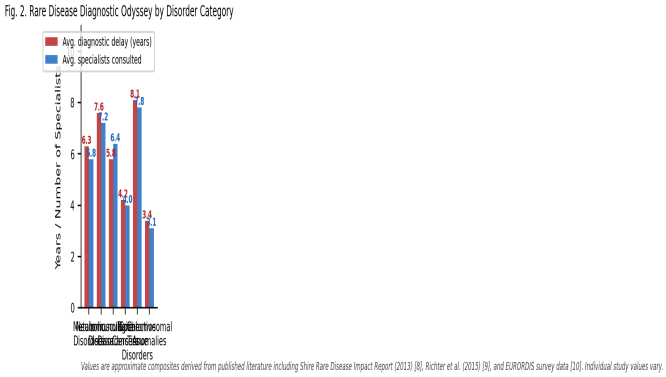

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Fig. 2. Approximate rare disease diagnostic delay and specialist burden by disorder category. Composite estimates derived from published literature [8][9][10]. Individual study values vary by methodology and population.

Across approximately 7,000 recognized rare diseases [8], the majority lack dedicated computational diagnostic support tools. HqA is designed to address this gap through phenotype-driven differential generation: given a patient's HPO-coded phenotype profile, the system computes semantic similarity scores against known disease phenotype signatures, ranks candidate diagnoses, and identifies the genomic variants and molecular network perturbations most consistent with each candidate. The result is a ranked differential for clinician review — not an autonomous diagnosis.

D. Clinical Decision Support Positioning

HqA is explicitly designed as a clinical decision support tool operating under physician authority, not an autonomous diagnostic system. All outputs are framed as ranked hypotheses for clinician evaluation, with evidence chains showing the data sources, variant annotations, and network evidence supporting each hypothesis. This design choice is grounded both in the current regulatory landscape for clinical AI and in the fundamental principle that clinical judgment incorporates contextual, relational, and ethical dimensions that computational systems do not capture.

IV. MNTU: Molecular Node Tracer Unified

A. Molecular Network Analysis Framework

MNTU is designed as a companion platform to HqA, providing deep molecular network analysis of disease mechanisms identified at the causation mapping layer. The system operates on three primary network types: protein-protein interaction (PPI) networks sourced from the STRING database [2] and the Human Protein Atlas [17]; metabolic pathway networks from the KEGG database [3]; and gene regulatory networks incorporating transcription factor binding and chromatin accessibility data.

B. Network Propagation and Disease Module Identification

MNTU implements network propagation algorithms — analogous to random walk with restart (RWR) approaches — to identify disease-relevant subnetworks from seed gene sets identified by HqA variant analysis. The theoretical basis is the disease module hypothesis of Menche et al. [11]: disease genes cluster in network neighborhoods, and their propagation patterns in the interactome reflect the mechanistic scope of the disease process. MNTU traces these propagation patterns to identify candidate therapeutic targets, secondary pathway involvement, and potential disease-disease mechanistic overlaps relevant to comorbidity understanding.

C. Graph Neural Networks for Molecular Analysis

For molecular property prediction and drug-target interaction analysis, MNTU incorporates graph neural network (GNN) approaches following the framework described by Wieder et al. [18] and the antibiotic discovery demonstration of Stokes et al. [19], who used a GNN trained on molecular graphs to identify a structurally novel antibiotic active against drug-resistant *Mycobacterium tuberculosis*. This approach represents the application of machine learning to molecular graph structure rather than sequence or feature vectors, capturing topological properties of molecular interaction relevant to biological function.

D. Protein Structure Integration

The availability of the AlphaFold protein structure database [20], containing structural predictions for over 200 million proteins, fundamentally changes the scope of structure-based analysis available to MNTU. Where previously structure-based drug discovery and protein interaction analysis required experimental structure determination for each target, AlphaFold predictions provide computationally accessible structural hypotheses for the full proteome. MNTU incorporates AlphaFold structural data as an additional evidence layer for protein interaction plausibility assessment and binding site identification.

V. Antimicrobial Resistance Surveillance

Antimicrobial resistance represents one of the most severe threats to global public health. The O'Neill Review on Antimicrobial Resistance [15], commissioned by the UK government, projected 10 million annual deaths from drug-resistant infections by 2050 absent coordinated global action — exceeding current annual cancer mortality. The WHO Global Antimicrobial Resistance and Use Surveillance System (GLASS) [16] documents substantial gaps in genomic resistance gene surveillance coverage across low- and middle-income countries. Fig. 3 illustrates the projected mortality trajectory and regional surveillance coverage gaps.

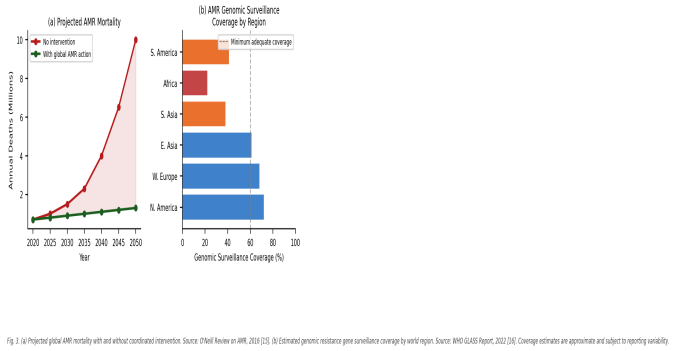

Fig. 3. (a) Projected global AMR mortality with and without coordinated intervention. Source: O'Neill Review [15]. (b) Estimated AMR genomic surveillance coverage by region. Source: WHO GLASS [16]. Values approximate.

HqA's AMR module is designed to integrate resistance gene identification — using reference databases such as ResFinder [21] and CARD (Comprehensive Antibiotic Resistance Database) — with epidemiological spread modeling and clinical outcome correlation. The computational goal is early identification of resistance gene emergence before clinical outbreak conditions, providing actionable signal to public health authorities within the surveillance-to-response latency window that determines containment feasibility.

Wastewater-based epidemiology, demonstrated by Peccia et al. [26] during the SARS-CoV-2 pandemic to track community infection dynamics ahead of clinical case reporting, represents a complementary surveillance modality for AMR monitoring. The integration of wastewater resistance gene sequencing with clinical genomic surveillance represents a planned extension of the HqA AMR module.

VI. Privacy-Preserving Health Research Architecture

A. The Health Data Silo Problem

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The development of effective computational disease models requires access to patient data at scale. The tension between this requirement and the privacy rights of patients — protected in Canada under the Personal Information Protection and Electronic Documents Act (PIPEDA) and provincial health privacy legislation — has historically resulted in data siloing that limits the statistical power and generalizability of computational health research. Bonomi et al. [25] document the specific privacy challenges attendant to genomic data sharing, noting that genomic data is inherently re-identifying and that current de-identification techniques provide incomplete protection against re-identification attacks.

B. Federated Learning

Federated learning, introduced by McMahan et al. [22] and applied to healthcare contexts by Rieke et al. [23] and Sheller et al. [24], provides a framework for training machine learning models across multiple data-holding institutions without centralizing the underlying patient data. In the federated paradigm, each institution trains a local model update on its own data; only the model gradients — not the patient records — are transmitted to a central aggregation server. The global model is updated from the aggregated gradients and redistributed. Sheller et al. [24] demonstrated that federated learning across distinct institutions can achieve model performance within 0.6% of centralized training on brain tumor segmentation tasks, establishing its clinical viability.

C. Zero-Knowledge Proof Integration

HqA and MNTU extend the federated learning framework with zero-knowledge proofs [25] from the PTH Meridian security stack (AKR Naos). ZKP allows an institution to prove that its local model update was computed from data satisfying specified properties — minimum cohort size, data quality thresholds, diagnostic category membership — without revealing the underlying data or the model gradients in full. This provides a mathematically verifiable privacy guarantee beyond the operational assurance provided by federated learning alone. Fig. 4 illustrates the combined architecture.

Fig. 4. Proposed privacy-preserving architecture integrating federated learning [22][23] with zero-knowledge proofs from AKR Naos. Patient data never leaves the institutional boundary. PIPEDA compliant by design.

VII. Health Equity and Underserved Populations

A. Indigenous Health in Canada

Indigenous peoples in Canada experience substantially worse health outcomes across virtually every measurable indicator, reflecting the long-term health consequences of colonization, displacement, and systemic inequities in healthcare access and quality. The First Nations Information Governance Centre [27] documents that First Nations adults on-reserve experience rates of diabetes, tuberculosis, and mental health disorders significantly exceeding national averages. Life expectancy gaps between Indigenous and non-Indigenous Canadians, while narrowing over recent decades, remain substantial.

HqA is designed from the outset to be deployable in resource-constrained settings including remote and rural communities, with explicit attention to the diseases of highest prevalence in Indigenous populations. The computational tools for rare disease diagnosis and AMR surveillance are of particular relevance to communities with limited specialist access and elevated infectious disease burden. Health equity is a design requirement, not an afterthought.

B. Energy Poverty and Health

Energy access is a determinant of health outcomes. The WHO estimates that household air pollution from cooking with solid fuels causes approximately 3.2 million deaths annually [28], predominantly in low- and middle-income countries. Absence of reliable electricity eliminates cold-chain capacity for vaccine storage, eliminates refrigeration for insulin and temperature-sensitive medications, and severely constrains the operation of medical equipment. The Global Burden of Disease study [29] identifies energy poverty as a modifiable risk factor for a broad range of preventable health outcomes.

PTH Meridian's Energy Division and Bioinformatics Division share a common mission in this domain: the Energy Division builds toward lower-cost distributed energy infrastructure; the Bioinformatics Division quantifies the health return on energy investment, creating the evidence base for infrastructure prioritization decisions. HqA's causal modeling capabilities include environmental exposure as a data layer, enabling analysis of energy-health causal pathways at the population level.

C. Pandemic Preparedness and Early Warning

The SARS-CoV-2 pandemic demonstrated that global spread can outpace detection by weeks, converting containable outbreaks into global emergencies. Peccia et al. [26] demonstrated that SARS-CoV-2 RNA concentrations in wastewater predicted clinical case counts four to ten days in advance in the New Haven, Connecticut catchment, providing an early warning signal preceding clinical case detection. The integration of wastewater epidemiology, genomic zoonotic spillover prediction, and global mobility-adjusted transmission modeling represents the computational infrastructure for a genuine pandemic early warning system. HqA's epidemiological data ingestion layer is designed to support this integration.

VIII. Ethical Framework

A. Open Source and Auditability

Clinical decision support tools that affect patient care have an accountability obligation that proprietary systems cannot satisfy by design. HqA and MNTU are published as open-source implementations — allowing clinicians, researchers, regulators, and patients to inspect the methods underlying any recommendation the systems generate. This is a non-negotiable design commitment grounded in the established principles of research reproducibility and clinical AI accountability.

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B. Human Clinical Authority

No output from HqA or MNTU constitutes a clinical diagnosis or treatment recommendation. All outputs are framed as evidence-supported hypotheses for physician evaluation. The system is designed to extend clinical reach — enabling a general practitioner in a remote setting to access rare disease diagnostic support that would otherwise require specialist referral — without removing clinical judgment from the decision pathway. The physician remains the principal. The system is the analyst.

C. Limitation Disclosure

HqA and MNTU are in active development and have not undergone clinical validation. No performance benchmarks against clinical populations are claimed in this paper. The methodological foundations described here are drawn from validated published approaches; their integration in the HqA and MNTU implementations requires independent clinical evaluation before deployment in patient care settings. This disclosure is maintained publicly in all PTH Meridian documentation and constitutes an unconditional ethical obligation.

D. Data Governance and PIPEDA Compliance

The federated ZKP architecture described in Section VI is designed to satisfy PIPEDA requirements by ensuring that no personally identifiable health information leaves the institutional custody of the data holder. Model gradients transmitted in the federated protocol are aggregated before any party outside the originating institution can access them. ZKP proofs provide verifiable assurance of data property compliance without data exposure. Institutional data sharing agreements, ethics board review, and patient consent frameworks remain the responsibility of participating institutions and are prerequisites for any deployment.

IX. Discussion

The computational health gap is not primarily a technological problem. The methods underlying HqA and MNTU — causal inference, phenotype ontology matching, network propagation, federated learning, graph neural networks — are established in the published literature. The gap is integrative and institutional: these methods exist in separate research silos, implemented in non-interoperable tools, and inaccessible to the clinical settings where they would have the greatest impact on patient outcomes.

HqA and MNTU are designed to close this integrative gap in the Canadian health research context first, with architecture designed for broader deployment. The specific focus on rare disease, AMR surveillance, Indigenous health, and pandemic preparedness reflects the Canadian epidemiological and equity context: populations and conditions that are underserved by existing computational tools and where the potential clinical impact is highest.

Several limitations of the current proposal merit acknowledgment. First, causal inference from observational health data requires careful control for confounding, and the causal claims HqA generates are hypotheses requiring clinical validation, not established causes. Second, the federated ZKP architecture introduces computational overhead relative to centralized training, and the performance-privacy tradeoff requires empirical characterization in health-specific deployment contexts. Third, the equity commitments described in Section VII require active engagement with Indigenous communities and remote health authorities as genuine partners in design, not merely as beneficiaries of externally designed tools. These are acknowledged obligations, not resolved challenges.

X. Conclusion

This paper has described the proposed architecture and ethical framework of HqA and MNTU: two complementary open-source computational health tools addressing the disease causation mapping and molecular network analysis problem space. The systems integrate causal inference methodology, genomic variant analysis, HPO-based phenotype matching, STRING-based protein interaction network analysis, KEGG metabolic pathway mapping, AlphaFold structural data, graph neural network molecular analysis, federated learning, and zero-knowledge proof privacy guarantees into a coherent, open-source, PIPEDA-aligned computational health research platform.

The six problem domains addressed — rare disease diagnostic delay, AMR surveillance gaps, health data privacy, Indigenous health equity, energy poverty and health, and pandemic early warning — represent areas where the computational infrastructure to translate available data into clinical and public health benefit is demonstrably insufficient. HqA and MNTU are proposed as contributions to closing this gap, beginning in the Canadian health research context and designed for broader deployment.

Clinical validation through institutional collaboration is the essential next step. PTH Meridian actively seeks partnerships with Canadian academic health centres, national research institutes, and Indigenous health organizations for collaborative validation studies. All implementations are published at github.com/PTHMeridian.

Glossary of Terms

AlphaFold

An AI system developed by DeepMind [20] that predicts three-dimensional protein structure from amino acid sequence with high accuracy. The AlphaFold Protein Structure Database contains predictions for over 200 million proteins.

Antimicrobial Resistance (AMR)

The ability of microorganisms to resist the effects of antimicrobial agents to which they were previously susceptible. The O'Neill Review [15] projects 10 million annual deaths from AMR by 2050 without coordinated global action.

Causal Inference

A statistical and mathematical framework for identifying causal rather than merely correlational relationships between variables. Founded by Pearl [4] using structural causal models and DAGs.

Directed Acyclic Graph (DAG)

A mathematical graph structure in which edges have a defined direction and no cycles exist. DAGs are used in causal inference to represent the assumed causal relationships among variables in a system.

Federated Learning

A distributed machine learning approach [22] in which model training occurs locally at each data-holding institution, with only model gradients — not patient data — transmitted to a central aggregator.

Human Phenotype Ontology (HPO)

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A controlled vocabulary of over 13,000 phenotypic abnormalities with explicit semantic relationships [12], enabling computational comparison of patient phenotype profiles against known disease signatures.

KEGG

Kyoto Encyclopedia of Genes and Genomes [3]: a database of metabolic, signaling, and regulatory pathway maps used as a reference for molecular network analysis in MNTU.

Molecular Node Tracer Unified (MNTU)

PTH Meridian's planned molecular network analysis platform for tracing disease origination and propagation across protein interaction, metabolic pathway, and gene regulatory networks.

PIPEDA

Canada's Personal Information Protection and Electronic Documents Act: federal legislation governing the collection, use, and disclosure of personal information in the course of commercial activity.

Protein-Protein Interaction (PPI) Network

A biological network in which nodes represent proteins and edges represent experimentally verified or computationally predicted interactions. The STRING database [2] is the primary PPI data source used in HqA and MNTU.

STRING

Search Tool for the Retrieval of Interacting Genes/Proteins [2]: a database of known and predicted protein-protein interactions, providing the PPI network foundation for HqA variant neighborhood analysis and MNTU network tracing.

Zero-Knowledge Proof (ZKP)

A cryptographic protocol in which a prover demonstrates a property of their data without revealing the data itself. In HqA/MNTU, ZKP allows institutions to verify data quality and cohort properties without data disclosure.

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